Orciny Device Animation Timeline

Version: V 0.9.0
Board: Adafruit Feather RP2040
Wings: Prop-Maker FeatherWing, 8-Channel Servo FeatherWing (PCA9685)

Animation Sequence Timeline

- **Program runtime:** Approximately 47 seconds from start of Boot Up to end of Device Failure
  - Performer is able to pause the sequence to slow it down or jump to Device Failure if it needs to finish faster

State 0: INACTIVE (Sequence 0)

- **0s+**: Sparse random single sparks every 3–5 seconds
- **Beam & NeoPixels:** Off

State 1: BOOT UP (Sequence 1)

- **0–4s**: Sparse sparks; cyan chases (slow to fast)
- **4–14s**: Sparks continue; cyan chases ramp up
- **14–22s**: Orange chases begin; pincers go rigid
- **22–24s**: Core pulses cyan/orange; beam fades to golden
- **24–26s**: Servo 1 oscillates
- **26–32s**: Both servos limp
- **32s+**: Transition to Demonstrate state

State 2: DEMONSTRATE (Sequence 2)

- **0s**: Core flashes cyan/orange; beam brightens to orange (90%)
- **0.5s**: Beam fades to teal; both servos rigid
- **1s**: Fluid pump and pulse filament start turning on and off at random intervals

- **1.5s**: Servos random movement (slow, then frantic)
- **2s**: Sparks start; magenta chases on core
- **3–7s**: Beam pulses teal/orange with accelerating frequency
- **7s**: Core all-off; beam turns magenta (30%)
- **7.25s**: Beam magenta fades to full; pincers freeze
- **9.5s**: Pincers freeze up
- **10.75s**: Transition to Device Failure state

State 3: DEVICE FAILURE (Sequence 3)

- **0s**: Random twitching servos; sparks increase in frequency; core rapid flashes all colors
- **0–4s**: Beam flashes teal/magenta/orange
- **1.75s**: Core → bright white
- **2s**: Core fades out; beam all colors full
- **2.5s**: Beam fades out; pincers go limp
- **3s**: Fluid pump seizes, pulse filament to full
- **3.5s**: Pincers go limp
- **4s+**: Everything off, output disabled

Notes

- **Physical Controls:**
  - SW1: Toggle all outputs on/off
  - SW2: Play/Pause
  - SW3: Jump to Device Failure
  - SW1 + SW3 held 5s: Reset to State 1, outputs off
- **See** `Orciny_Device.ino` for detailed implementation.